

PROJECT DESIGN PHASE-II

Technology Stack (Architecture & Stack)

Date: September 28, 2026	Team ID: SWTID-2026-9692
Project Name: AI WeatherWise API	Marks: 4 Marks

Technical Architecture Overview

The AI WeatherWise API is designed following a headless, decoupled MVC (Model-View-Controller) architecture pattern. Incoming HTTP requests pass through an Express Server Gateway operating on port 5001, where JWT Authentication Middleware enforces access controls. Business controller logic delegates persistence to MongoDB via Mongoose and orchestrates external inferences using Google Gemini AI models.

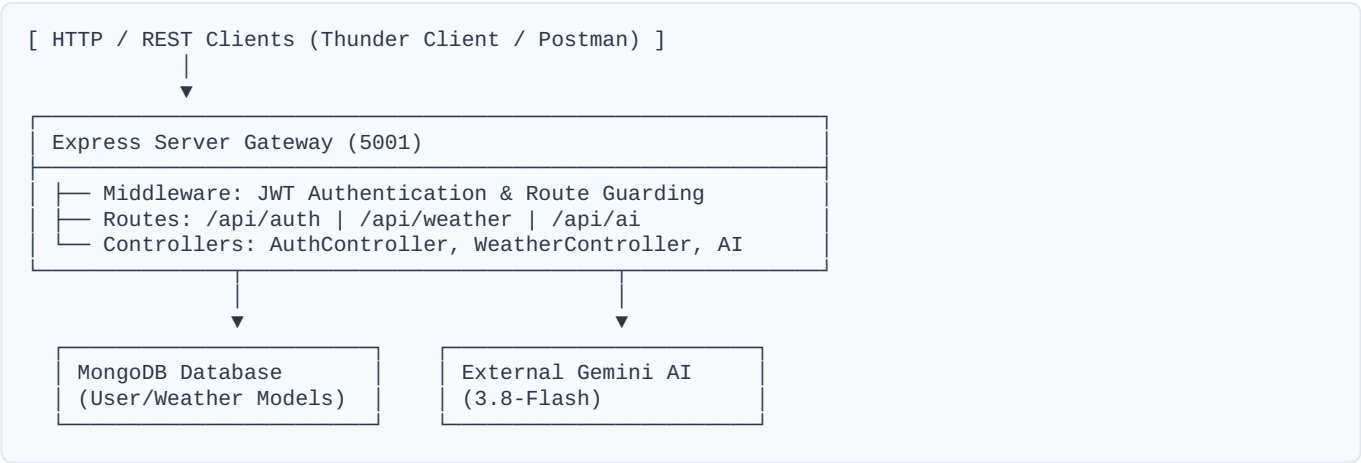


Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface / Client Interface	Headless RESTful API interface queried via API client tools and HTTP consumers.	Thunder Client, VS Code REST Client, JSON API
2	Application Logic - Gateway & Routing	Core server listener, express routing matrix, and JSON request body parsing.	Node.js (v20+), Express.js
3	Application Logic - Middleware	Stateless JWT token verification, route guarding, and CORS management.	jsonwebtoken (JWT), Cors
4	Application Logic - Controllers	Auth logic (register/login), weather data handlers, and AI prompt orchestration.	JavaScript (ES6 Node modules)
5	Database Layer	NoSQL document database storing registered user profiles and user credentials.	MongoDB (127.0.0.1:27017), Mongoose ODM
6	External API - Weather Service	Local/External endpoint providing city-level weather attributes (Chennai temp, humidity, condition).	REST API Endpoint (/api/weather/:city)
7	External API - AI Insights		

S.No	Component	Description	Technology
		Large Language Model API used for generating natural language weather summaries and advice.	Google Generative AI SDK (<code>@google/generative-ai</code>), <code>gemini-3.8-flash</code>
8	Infrastructure & Server	Local Node development execution server running continuously on port 5001.	Local Node Server (<code>npm run dev</code>), nodemon

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Leverages open-source server runtimes and database ODM wrappers for rapid development.	Node.js, Express.js, Mongoose ODM
2	Security Implementations	Bcrypt password hashing, stateless JWT Bearer token evaluation, and NoSQL injection safeguards.	bcryptjs, jsonwebtoken, CORS Header Guard
3	Scalable Architecture	Decoupled MVC architecture separating routing, controller logic, and database persistence.	RESTful Microservices Architecture Pattern
4	Availability	Persistent database connection loop with MongoDB daemon and auto-restarting Node process.	MongoDB Native Driver, Nodemon Process Supervisor
5	Performance	Sub-200ms local execution for auth routes and sub-2.5s for asynchronous external AI inference execution.	Asynchronous Non-blocking Event Loop (Node.js)

References

- <https://c4model.com/>
- <https://expressjs.com/>
- <https://mongoosejs.com/>
- <https://ai.google.dev/gemini-api/docs>